TBK

Bavariaring 4-6, 80336 München

TBK, Bavariaring 4-6, 80336 München

An das
Europäische Patentamt

80298 München

May 20, 2025

European Patent Application No.: 22 880 631.1-1001
Applicant: SUMITOMO ELECTRIC INDUSTRIES, LTD.
TBK ref.: EP 110150

Reference is made to the Official Communication dated January 10, 2025, and to the Official Communication dated December 17, 2024, comprising the Extended European Search Report.

The applicant intends to proceed further with this application.

For the examination proceedings herewith

- a new set of claims 1 to 8, and
- new description pages 1 to 23

are filed to replace the claims and the corresponding description pages as filed when entering the regional phase before the European Patent Office.

The claims have been completed with reference signs in brackets.

Claim 1 has been put into the two-part form.

Patentanwälte
Aurel Vollnhals, Dipl.-Ing.
Thomas J.A. Leson, Dipl.-Ing.
Matthias Grill, Dipl.-Ing.
Rainer Böckelen, Dipl.-Ing.
Stefan Klingele, Dipl.-Ing.
Jürgen Faller, Dipl.-Ing.
Jürgen Tefert, Dipl.-Ing.
Susanne Popp, Dipl.-Ing.
Ralf Brockhaus, Dipl.-Ing.
Markus Rossel, Dipl.-Ing.
Roland Lang, Dipl.-Ing.
Tobias Schob, Dipl.-Ing.
Jörg Preuß, Dipl.-Ing.
Gernot Pirker, Dr., Dipl.-Ing.
Christoph Schmidt, Dipl.-Ing.
Christoph Birghan, Dr., Dipl.-Biol.
Michael Hutter, Dipl.-Ing.
Andreas Wagner, Dipl.-Ing.
Peter Tautscher, Dipl.-Ing.
Wolfgang Wittmann, Dipl.-Ing.
Gregor Michels, Dr., Dipl.-Phys.
Andreas Enthart, Dr., Dipl.-Chem.
Timo Huber, Dr., Dipl.-Chem.
Florian Manzenrieder, Dr., Dipl.-Chem.
Marco Bauer, Dipl.-Ing.
Stefan Dillis, Dipl.-Ing.

Rechtsanwälte
Maximilian Rackl

Japanische Patentanwälte
日本国弁理士
Yuki Inoue, 井上 悠輝, BSc
Tomoko Tomioka, 富岡 知子, MEng
Sayaka Tomasi, 戸崎 沙也夏, BA

T: + 49. 89. 544 69 0
F: + 49. 89. 544 69 290
tbkpost@tbk.com
tbk.com

A03030596 6
GD263/PG155/VA112
/Instruktionen für Draft bearbeiten

Claim 8 has been added to the claim set.

The subject-matter of claim 8 disclosed in paragraphs [0044] and [0045].

Moreover, the subject-matter of dependent claim 8 is admissible. Although one might determine that the wording of claim 8 defines the embodiment by a result to be achieved, it has to be admitted that for performing the subject-matter of claim 8, the skilled person would not need undue experimentation. That is, it can be easily determined whether the features of claim 8 are met or not.

Regarding claim 1, the Search Examiner's opinion is respectfully contradicted to. That is, the subject-matter of claim 1 is not disclosed in D1.

In the Examiner's opinion, D1 discloses a reference surface 34 (see Fig. 6) and contact parts in the form of the corners of the octagon (not hexagon as determined in the ESO) visible in Fig. 6.

While surface 34 of D1 might indeed be a reference surface, claim 1 clearly defines a reference surface having two contact parts. Even when considering surface 34 of D1 as reference surface, it merely has one contact part. In D1, each corner of the octagon forms a contact part. In Fig. 6, there are four of the corners in contact with the flange's corners. However, while each one of the eight corners even belongs to two adjacent contact surfaces, it is obvious from Fig. 6 of D1 that each of the reference surfaces cannot have two contact parts. That is, only each second corner is in contact with the flange's corners. Thus, every second corner is not a contact part. Therefore, D1 fails to disclose a reference surface having two contact parts.

Moreover, the Examiner considers Figs. 1 and 2 of D2 to anticipate the subject-matter of claim 1.

However, from these figures, it cannot at all be said in what manner, if at all, there is contact between outer peripheral surface of the flange and an inner surface of the housing. In Fig. 2, the only contact among the flange and the

housing is at the left side of the flange where it appears to abut against a transverse wall (40) of the housing.

Regarding D3, the Search Examiner's statement lacks accurate disclosure indication.

Moreover, our analysis reveals that the features claimed in claim 1 are not disclosed in Figs. 4 and 5. Some embodiments (see Figs. 17 to 19) appear to disclose an arrangement similar to that of D1. However, as in D1, there is no disclosure of one reference surface having two contact parts.

Therefore, the subject-matter of claim 1 is new.

Moreover, the subject-matter of claim 1 is based on an inventive step. The difference between the subject-matter of claim 1 and closest prior art document D1 are the features in the characterizing portion.

These features enable reference for rotation angle of the flange. Moreover, a rotational direction of the optical fiber having the ferrule attached to the tip portion is also adjusted based on the reference for rotation angle of the flange. At this time, the optical fiber is aligned with the flange. With this configuration, the rotational direction of the flange in the housing is fixed, and as a result, a variation in an alignment state of the optical fiber is effectively suppressed.

Such an effect cannot at all be obtained by any of the prior art documents, not at all to mention that they of course do not even incite such an effect or the solution to obtain the same.

Correspondingly, the subject-matter of claim 1 is based on an inventive step.

Regarding claim 3, it is noted that D2 altogether has only 7 figures. Thus, the Examiner's statement that Fig. 17 of D2 discloses the features of claim 3 cannot be followed.

The description has been adapted by incorporating mention of D1 and D2. The term "disclosure" has been replaced by "invention". Regarding item 4.3, the objection is not understood. None of the cited paragraphs discloses any claim-like statement suitable to render the subject-matter of the claims unclear. This even more, since according to jurisdiction, the scope of protection is defined by the claims and not by the description. The mentioned paragraphs [0001] to [0018] pertain to the field of the invention, the background art, the summary of the invention which corresponds to claim 1, the brief description of the drawings, and a general description of embodiments corresponding to claims 1 to 7 together with advantages thereof.

Such a general description of embodiments corresponding to the claims is part of almost any European Patent Application. The wording of the general description describing the embodiments corresponds to the wording of the claims. Therefore, it is not understood, how this wording should cause unclear matter.

Thus, should the Examiner not follow the above, the Examining Division is kindly asked to specify, which claim-like statements in paragraphs [0001] to [0018] they identify suitable to render the claims themselves unclear.

It is believed that the documents now on file are sufficient for the grant of a patent, the communication under Rule 71(3) EPC being expected as next step of the proceedings.

However, should the Primary Examiner consider it to be necessary to further discuss the matter of the application before issuing another communication, the representative is available for a personal consultation by telephone or in the course of an informal interview.

Oral Proceedings are requested as an auxiliary measure.

Dr. Gernot Pirker
Patentanwalt
TBK

Enclosures:
- New claims 1 to 8 (clean copy and working copy)
- New description pages 1 – 23 (clean copy)
- New description pages 1 – 24 (working copy)